

$$Pr(y|\lambda) = \frac{\lambda^y e^{-\lambda}}{y!}$$

$$L[\phi] = -\log [Pr(y|f[x, \phi])] = -\log \left[\frac{\lambda^y e^{-\lambda}}{y!} \right]$$

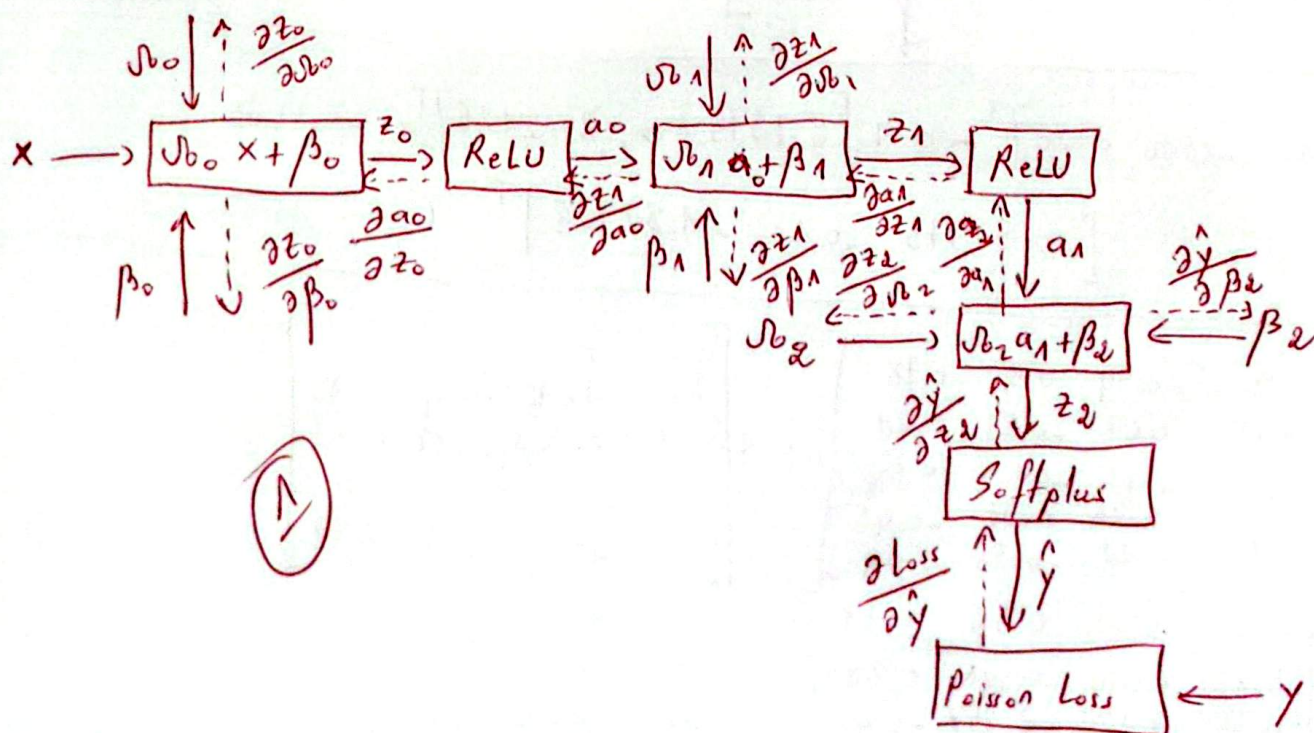
$$= - \left[\log(\lambda^y e^{-\lambda}) - \log(y!) \right] = -\log(\lambda^y e^{-\lambda}) + \log(y!)$$

$$= -\log \lambda^y - \log e^{-\lambda} = -y \log \lambda + \lambda = \hat{y} - y \log \hat{y}$$

→ constant

$$\beta_0^{t+1} = [0.01, -0.02, 0.03, -0.04, 0.05] - 0.1 [0, -3.44, 19.27, 0, 12.35]$$

$$= [0.01, 0.324, -1.897, -0.04, -1.185]$$



$$\mathbf{w}_2^t = [-0.25, 0.50, -0.75]$$

$$\nabla \mathbf{w}_2^t = [-16.506, 0, -49.843]$$

$$\mathbf{w}_2^{t+1} = [1.4006, 0.50, 4.2343]$$

$$\phi^{t+1} \leftarrow \phi^t - \alpha \nabla \phi^t$$

$$\alpha = 0.1$$

$$\beta_2^t = [0.05]$$

$$\nabla \beta_2^t = [-26.95]$$

$$\beta_2^{t+1} = [2.745]$$

$$\begin{aligned} \mathbf{w}_{b1}^{t+1} &= \begin{bmatrix} 0.23 & -0.45 & 0.67 & -0.89 & 0.11 \\ -0.12 & 0.34 & -0.56 & 0.78 & -0.9 \\ 0.91 & -0.02 & 0.73 & -0.64 & 0.55 \end{bmatrix} \begin{bmatrix} 0 & 10.86 & 11.58 & 0 & 5.22 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 32.51 & 34.74 & 0 & 15.67 \end{bmatrix} \\ &= \begin{bmatrix} 0.23 & -1.536 & -0.488 & -0.89 & -0.412 \\ -0.12 & 0.34 & -0.56 & 0.78 & -0.9 \\ 0.91 & -3.271 & -2.744 & -0.64 & -1.017 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \beta_1^{t+1} &= [0.10, -0.10, 0.20]^T - 0.1 [6.7375, 0, 20.2125]^T \\ &= [-0.57375, -0.10, -1.82125]^T \end{aligned}$$

$$\begin{aligned} \mathbf{w}_0^{t+1} &= \begin{bmatrix} 0.12 & -0.34 & 0.56 & 0.78 \\ -0.45 & 0.67 & -0.89 & 0.21 \\ 0.31 & 0.42 & -0.53 & 0.64 \\ 0.15 & -0.26 & 0.37 & -0.48 \\ -0.11 & 0.22 & -0.33 & 0.44 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ -2.09 & -13.76 & -3.44 & -1.89 \\ 11.73 & 77.08 & 19.27 & 10.60 \\ 0 & 0 & 0 & 0 \\ 7.51 & 49.43 & 12.35 & 6.79 \end{bmatrix} \\ &= \begin{bmatrix} 0.12 & -0.34 & 0.56 & 0.78 \\ -0.241 & 2.046 & -0.546 & 0.399 \\ -0.863 & 7.288 & -2.457 & -0.42 \\ 0.15 & -0.26 & 0.37 & -0.48 \\ -0.861 & -4.723 & -1.565 & -0.239 \end{bmatrix} \end{aligned}$$

$$x = [0.608, 4, 1, 0.550]^T$$

$$B_0 = \begin{bmatrix} 0.12 & -0.34 & 0.56 & 0.78 \\ -0.45 & 0.67 & -0.89 & 0.21 \\ 0.31 & 0.42 & -0.53 & 0.64 \\ 0.15 & -0.26 & 0.37 & -0.48 \\ -0.11 & 0.22 & -0.33 & 0.44 \end{bmatrix}$$

$$\beta_0 = [0.01, -0.02, 0.03, -0.04, 0.05]^T$$

$$B_1 = \begin{bmatrix} 0.23 & -0.45 & 0.67 & -0.89 & 0.11 \\ -0.12 & 0.34 & -0.56 & 0.78 & -0.90 \\ 0.91 & -0.02 & 0.73 & -0.64 & 0.55 \end{bmatrix}$$

$$\beta_1 = [0.10, -0.10, 0.20]^T$$

$$B_2 = [-0.25, 0.50, -0.75] \quad \beta_2 = [0.05]$$

$$z_0 = [-0.28804, 1.6119, 1.72048, -0.8828, 0.77512]^T$$

$$a_0 = [0, 1.6119, 1.72048, 0, 0.77512]^T$$

$$z_1 = [0.61233, -1.21316, 1.84922]^T$$

$$a_1 = [0.61233, 0, 1.84922]^T$$

$$z_2 = -1.48999$$

Softplus function : $\text{softplus}(z) = \log(1 + e^z)$

$$\hat{y} = \log(1 + e^{-1.48999}) = 0.2035$$

$$L_{\text{poisson}}(y, \hat{y}) = \hat{y} - y \log(\hat{y})$$

$$L = 0.2035 - 30 \log(0.2035) = 47.96$$

$$L(\hat{y}, y) = \hat{y} - y \log(\hat{y})$$

$$\frac{\partial L}{\partial \hat{y}} = 1 - \frac{y}{\hat{y}}$$

0.5

$$y = 30, \hat{y} = 0.2035 \Rightarrow \frac{\partial L}{\partial \hat{y}} = 1 - \frac{30}{0.2035} = -146.43$$

$$\hat{y} = \log(1 + e^{z_2})$$

$$\frac{\partial \hat{y}}{\partial z_2} = \frac{e^{z_2}}{1 + e^{z_2}} = \sigma(z_2)$$

0.5

$$z_2 = -1.48999 \Rightarrow \frac{\partial \hat{y}}{\partial z_2} = \frac{e^{-1.48999}}{1 + e^{-1.48999}} = 0.184$$

$$z_2 = \beta_2 a_1 + \beta_2$$

$$\frac{\partial L}{\partial \beta_2} = \frac{\partial z_2}{\partial \beta_2} \cdot \frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} \cdot a_1^T$$

$$= -146.43 \times 0.184 \times [0.61233, 0, 1.84922]$$

$$= [-16.506, 0, -49.843]$$

0.5

$$\frac{\partial L}{\partial \beta_2} = \frac{\partial z_2}{\partial \beta_2} \cdot \frac{\partial L}{\partial z_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} = -146.43 \times 0.184 = -26.95$$

$$\frac{\partial L}{\partial a_1} = \frac{\partial z_2}{\partial a_1} \cdot \frac{\partial L}{\partial z_2} = \beta_2^T \cdot \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_2} = [-0.25, 0.50, -0.75]^T \times (-26.95)$$

$$= [6.7375, -13.475, 20.2125]^T$$

$$a_1 = \text{ReLU}(z_1)$$

0.6

$$\frac{\partial L}{\partial z_1} = \frac{\partial a_1}{\partial z_1} \odot \frac{\partial L}{\partial a_1} = [1, 0, 1]^T \odot [6.7375, -13.475, 20.2125]^T$$

$$= [6.7375, 0, 20.2125]^T$$

$$z_1 = \omega_{b1} a_0 + \beta_1$$

$$\frac{\partial L}{\partial \omega_{b1}} = \frac{\partial z_1}{\partial \omega_{b1}} \cdot \frac{\partial L}{\partial z_1} = \frac{\partial L}{\partial z_1} \cdot a_0^T = [6.7375, 0, 20.2125]^T \cdot [0, 1.6119, 1.72048, 0, 0.77512]^T$$

$$= \begin{bmatrix} 6.7375 & 6.7375 & 6.7375 & 6.7375 & 6.7375 \\ 0 & 0 & 0 & 0 & 0 \\ 20.2125 & 20.2125 & 20.2125 & 20.2125 & 20.2125 \end{bmatrix}$$

$$[0, 1.6119, 1.72048, 0, 0.77512] = \begin{bmatrix} 0 & 10.86 & 11.58 & 0 & 5.22 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 32.51 & 34.74 & 0 & 15.67 \end{bmatrix}$$

$$z_1 = \omega_{b1} a_0 + \beta_1$$

0.5

$$\frac{\partial L}{\partial \beta_1} = \frac{\partial z_1}{\partial \beta_1} \cdot \frac{\partial L}{\partial z_1} = [6.7375, 0, 20.2125]^T$$

$$\frac{\partial L}{\partial z_0} = \frac{\partial a_0}{\partial z_0} \odot \frac{\partial L}{\partial a_0} = \frac{\partial L}{\partial z_0} \cdot \text{ReLU}'[z_0]$$

$$z_1 = \omega_{b1} a_0 + \beta_1$$

$$\frac{\partial L}{\partial a_0} = \frac{\partial z_1}{\partial a_0} \cdot \frac{\partial L}{\partial z_1} = \omega_{b1}^T \cdot \frac{\partial L}{\partial z_1} = \begin{bmatrix} 0.23 & -0.12 & 0.91 \\ -0.45 & 0.34 & -0.02 \\ 0.67 & -0.56 & 0.73 \\ -0.89 & 0.78 & -0.64 \\ 0.11 & -0.90 & 0.55 \end{bmatrix} [6.7375, 0, 20.2125]^T$$

$$= [19.92175, -3.440125, 19.27215, -18.931375, 12.358]^T$$

$$a_0 = ReLU(z_0)$$

$$\frac{\partial L}{\partial z_0} = \frac{\partial a_0}{\partial z_0} \odot \frac{\partial L}{\partial a_0} = [0, 1, 1, 0, 1]^T \cdot \begin{bmatrix} 19.92175, -3.440125, 19.27215, \\ -18.931375, 12.358 \end{bmatrix}^T$$

$$= [0, -3.440125, 19.27215, 0, 12.358]^T$$

0.5

$$z_0 = w_0 x + \beta_0$$

$$\frac{\partial L}{\partial w_0} = \frac{\partial z_0}{\partial w_0} \cdot \frac{\partial L}{\partial z_0} = \frac{\partial L}{\partial z_0} \cdot x^T =$$

$$[0, -3.440125, 19.27215, 0, 12.358]^T \cdot [0.608, 4, 1, 0.550] =$$

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ -3.440125 & \dots & -3.440125 \\ 19.27215 & \dots & 19.27215 \\ 0 & \dots & 0 \\ 12.358 & \dots & 12.358 \end{bmatrix} \cdot [0.608, 4, 1, 0.550] =$$

0.5

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ -2.09117 & -13.7605 & -3.440125 & -1.89206875 \\ 11.7320192 & 77.0886 & 19.27215 & 10.76036925 \\ 0 & 0 & 0 & 0 \\ 7.51424 & 49.432 & 12.358 & 6.79169 \end{bmatrix}$$

$$z_0 = w_0 x + \beta_0$$

$$\frac{\partial L}{\partial \beta_0} = \frac{\partial z_0}{\partial \beta_0} \cdot \frac{\partial L}{\partial z_0} = \frac{\partial L}{\partial z_0} = [0, -3.440125, 19.27215, 0, 12.358]^T$$

①

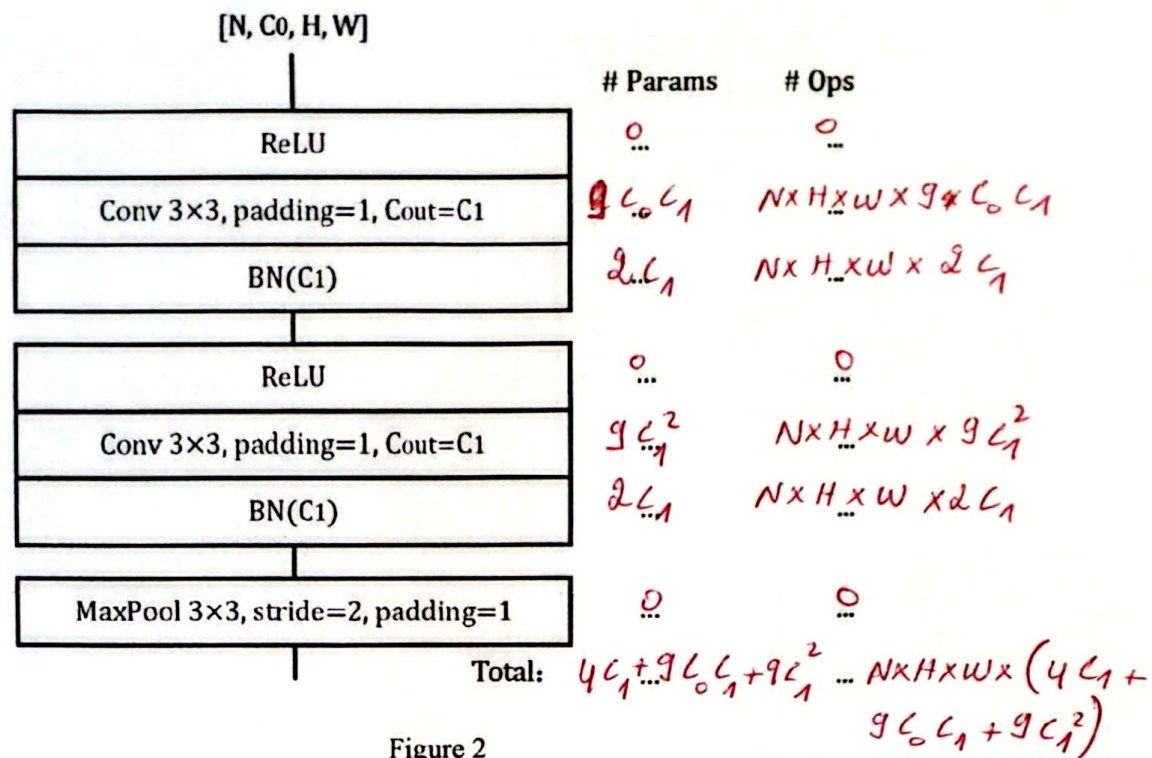


Figure 2

②

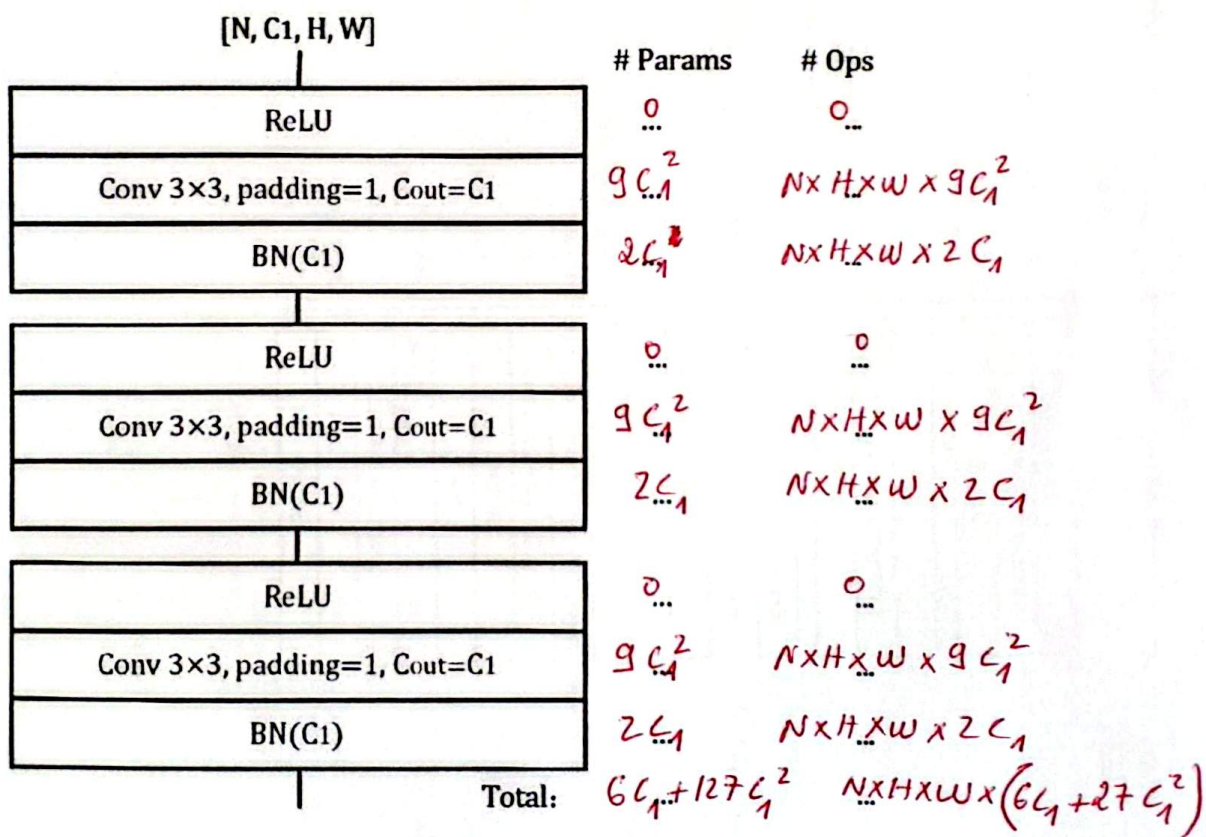


Figure 3

